Adaptive Optics
adapted to microscopy

a division of imagine optic

mu-DM User Manual

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Table of contents

1 Purpose of the document 1
2 Safety 1
3 Contents of package 1
4 Specifications 3
5 Mounting the mirror 4
6 Reflectivity 7
7 Hardware Setup 7
8 Software Installation: mu-DM Suite 8
9 Using mu-DM software 9
9.1 Starting and stopping the mirror 9
9.2 Applying shapes to the mirror 10
9.3 Message log 10
10 Using mu-DM with WaveTune 10
10.1 Error messages 11
10.2 Trigger 12
10.3 Full speed 13
10.4 Mirror temperature and flat 13
11 Precautions 14
12 Support 14

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1 Purpose of the document

This document details all specifications and information needed to mount and start using mu-DM. mu-DM is a deformable mirror with 91 electromagnetic actuators, with an optimized actuator layout.
It can be implemented as a standalone product in an open-loop or along with a wavefront sensor for closed-loop applications.

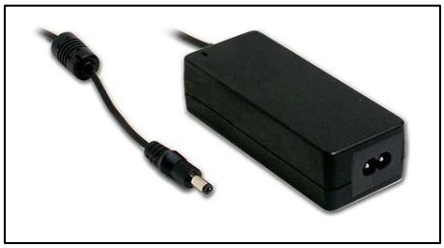
2 Safety

Do not disassemble the product or use it for anything else but its intended use.

3 Contents of package

Upon reception, the package contains:

mu-DM and its cover



mu-DM power supply and cable

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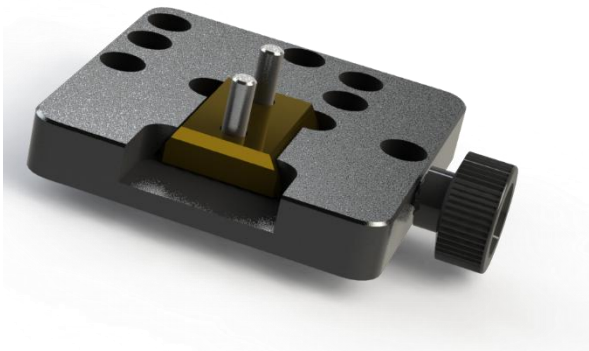
USB3 cable

Trigger cable



Mounting adapter and screws (screws are not mounted on the adapter upon reception)

USB key containing configuration files, software, and user manual

4 Specifications

Environmental conditions	
General	Indoor use
Working temperature	19-25°C
Working humidity	40-60% RH non-condensing
Storage temperature	15-30°C
Storage humidity	30-70% RH non-condensing

Deformable mirror	
Dimensions	93.8 x 98.3 x 67.2mm^3
Weight	250g
Pupil diameter	15mm
Number of actuators	91
Coating	Protected silver
Surface quality (deformation)	7nm RMS*
Maximum generated wavefront (PV)	1 actuator: > 10µm
	7 actuators: > 50µm
Temporal stability (open loop)**	<15nm RMS over 12h

*Optional: down to 4nm RMS

**Optional: if regulation option is activated

Electronics	
Main Input voltage	5V
Main input current	6A
Trigger	TTL signal (5V)
Connexion	USB 3.0

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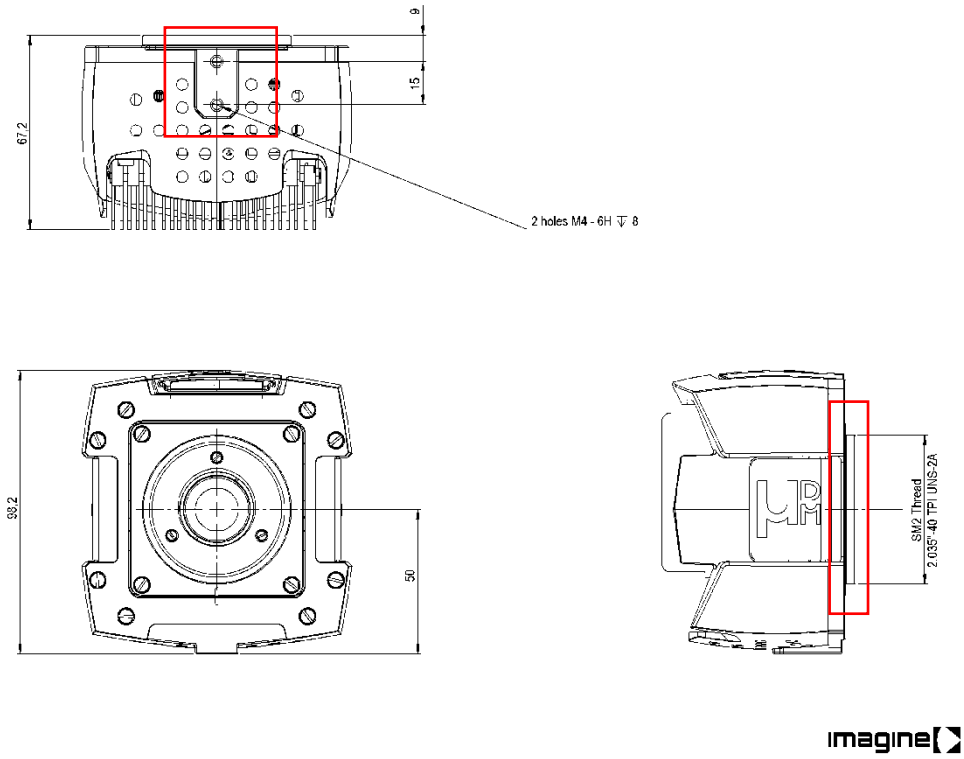
5 Mounting the mirror



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There are 2 ways of directly mounting the mirror on an optical bench:

- Using the two M4-6H holes at the base of the mirror
- using the SM2 thread at the front of the mirror

You can also set up the mirror using the included mounting adapter.
This interface can be used with several standard mounts, such as Thorlabs KM200PM/M, Optosigma kinematic rotation stage and Imagine Optic 5 axis stage.

The mirror weighs 250g, which can be on the limit of performance of some mounts. When aligning with a mount such as Thorlabs KM200PM/M, please make sure you do not push on the mirror in such a way that you slightly dislocate the mount into the wrong position.

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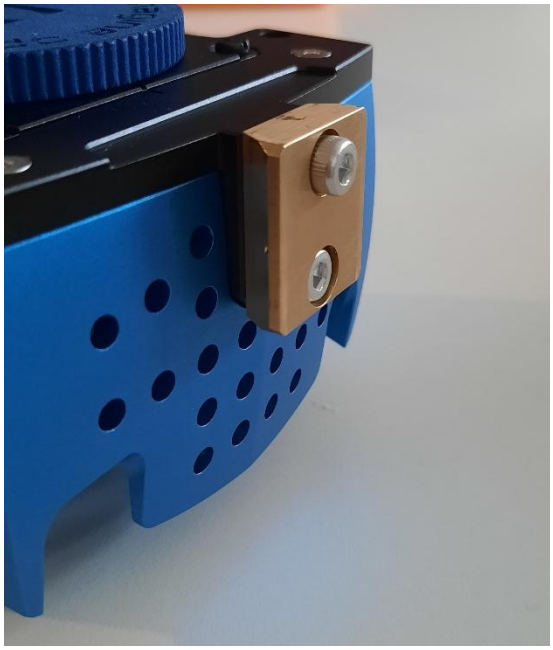
Thorlabs KM200PM/M

Optosigma kinematic rotation stage

Imagine Optic 5 axis tip/tilt/translation stage

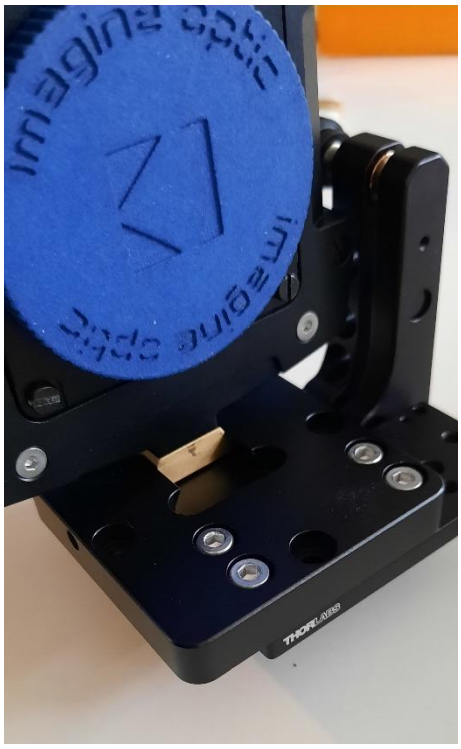
The following instructions explain how to mount the mirror using the adapter :

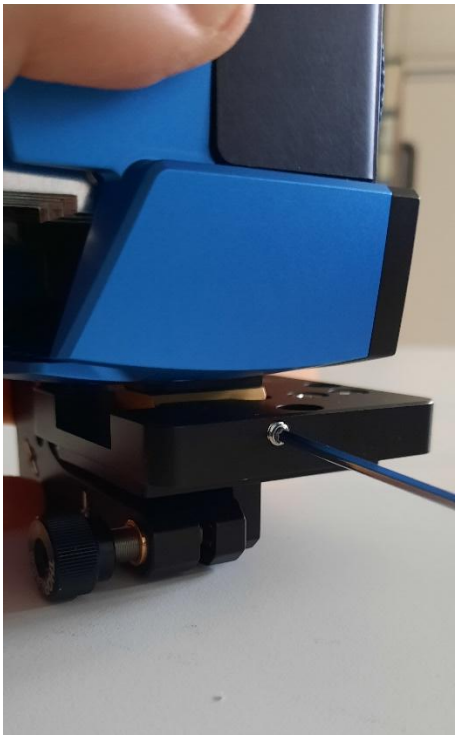
1- Mount the gold part of the adapter onto the base of the mirror using the two provided M4x8 screws:



2- Mount the black part of the adapter onto the desired mount. The following picture shows how to mount the adapter on the Thorlabs KM200PM/M mount using 4 M4x8 screws:

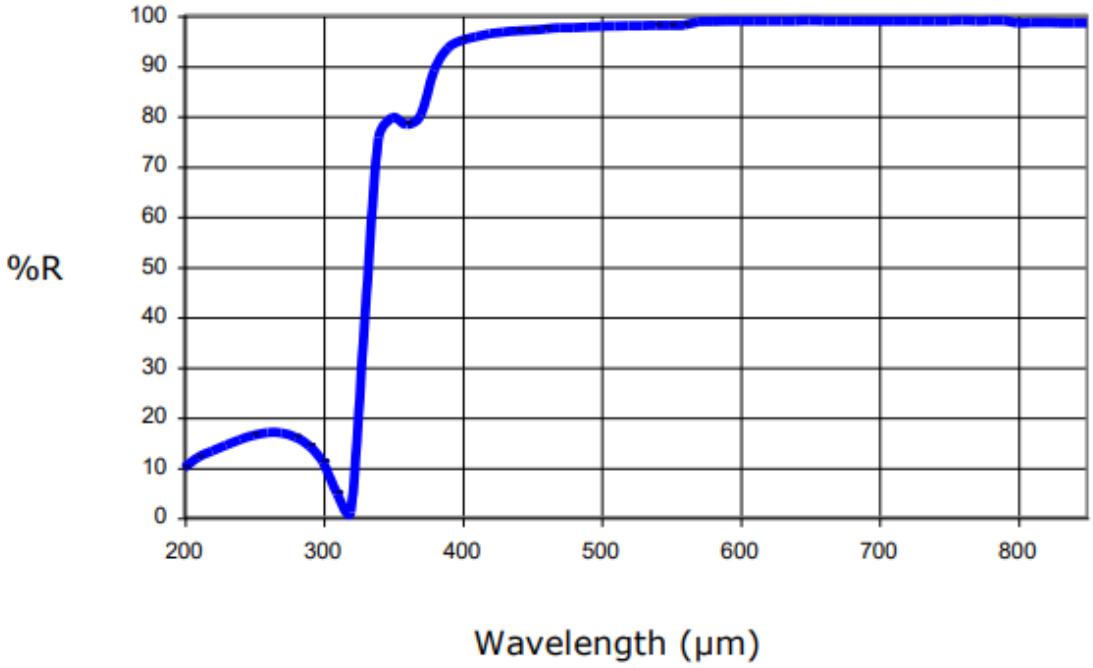
3- Slide the gold part of the adapter into the slot of the black part of the adapter, and fix it into place with the provided M4 screw mounted on the side of the adapter:





6 Reflectivity

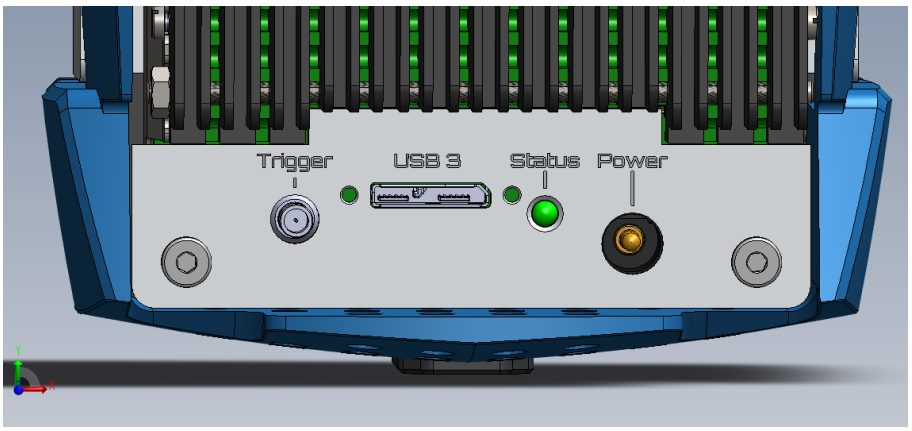
The membrane of the deformable mirror is coated with protected silver.



In front of the membrane is a protective window with antireflective coating in the visible spectrum, unless otherwise specified during purchase. Typical transmission is above 99.5% over 350-700nm.

7 Hardware Setup

To operate the mirror, connect the power supply to “POWER”.



Then plug the USB 3.0 cable to “USB 3”. Both cables must be connected to the computer to operate mu-DM.

There is no physical button to switch on the mirror. It is switched on through its control software.

The status LED will light up green upon connexion. In 2-5s it will switch off if all mirror components are working correctly. If the LED does not switch off, this means regulation is not activated. When an error appears, the LED turns orange.

If your mirror has the regulation option, we recommend waiting until the mirror temperature is stable before using the mirror (about 30-45 minutes).

8 Software Installation: mu-DM Suite

To install the mirror, please use the installer *MuDM Installer.exe* provided on the installation key.

Check that the driver has been installed correctly: device should be renamed *FTDI FT600 USB3.0 Bridge Device* in the *Device Manager*. If the driver is not installed, software will crash upon connection of the mirror.

If the mirror driver does not install correctly, please install drivers manually. To do this, go to *Device Manager* and find the device corresponding to the mirror (in *Universal Serial Bus controllers*). Right click on the device, then select *Update Driver.* Click on *Browse my computer for drivers,* point to the driver's folder on the USB key (*FTD3XXDriver_WHQLCertified_v1.3.0.4*) and make sure to include subfolders.

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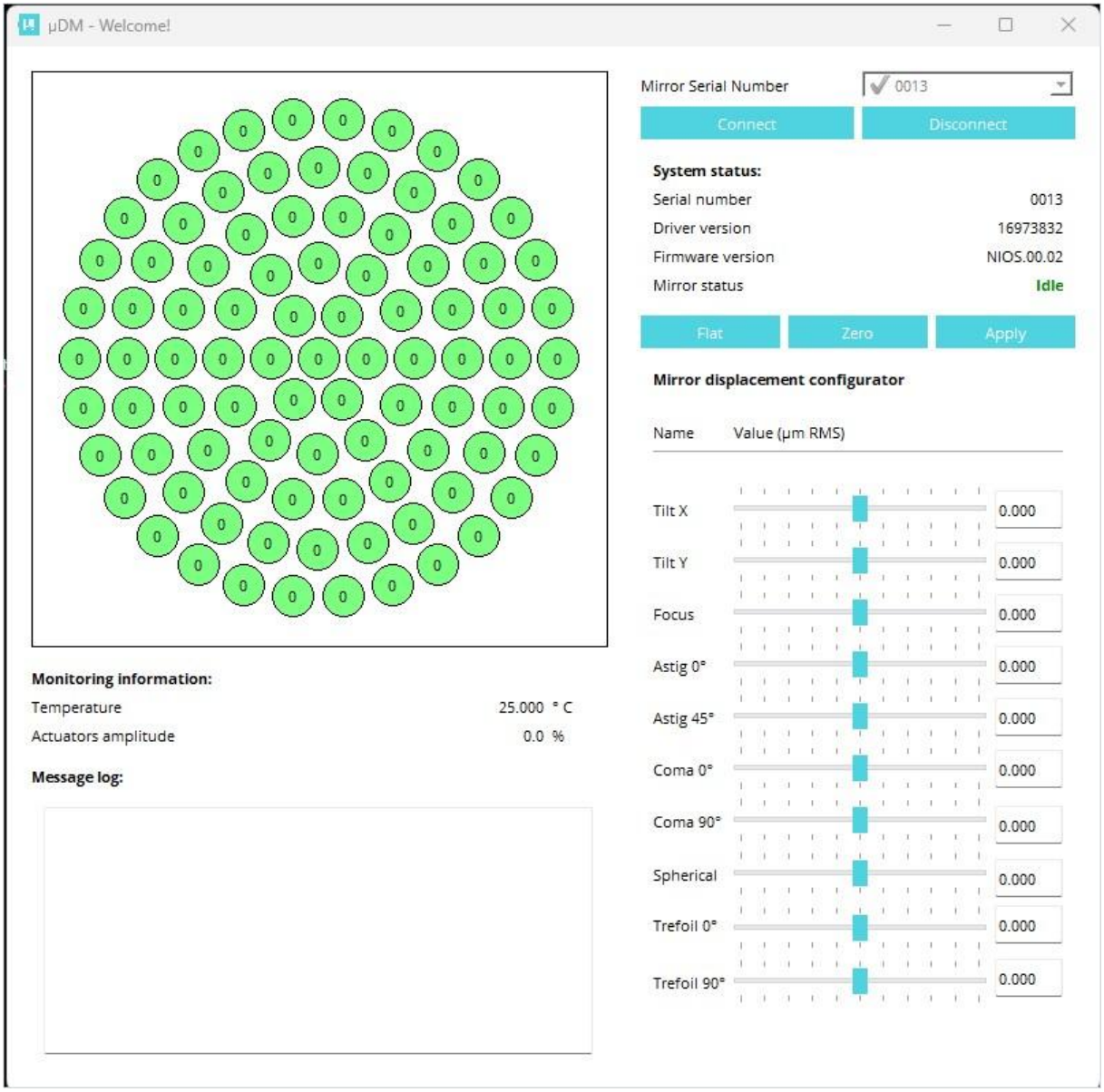
9 Using mu-DM software



9.1 Starting and stopping the mirror

If you have more than one mirror, choose the mirror you want to connect in the SN box. If you have only one mirror, the right serial number will appear in the box. A green checkmark beside the mirror serial number indicates the mirror has been detected. Then click on *Connect*. *Mirror status* should change from *Disconnected* to *Pending* while the mirror is connecting to *Idle* once the mirror is ready for use.

If the mirror does not connect, please make sure both the USB cable and the power supply are plugged in, and that the mirror driver has been correctly installed.

To stop the mirror, click on *Disconnect*. Mirror status will indicate *Disconnected* once the mirror is off.

9.2 Applying shapes to the mirror

Each actuator can be controlled with a current ranging from -500mA to 500mA. The layout shows each actuator and the amount of current that is applied to it.

Flat applies the factory setup shape that corresponding to a plane mirror.

Zero resets to zero the current applied to each actuator.

Use the sliders under *Mirror displacement configurator*, up/down arrows on keyboard, or type in the value to configure the combination of Zernike aberrations. Then click on *Apply*.

To apply voltage to a single actuator, click on the actuator in the layout, then use the slide bar that appears in *Mirror displacement configurator* to change the amount of current to send to the actuator and click on *Apply*. To go back to the Zernike aberration sliders, click on the blank space around the actuators on the actuator layout.

Actuators amplitude gives the percentage of current applied on the mirror over the total current that can be applied. If the total current exceeds 100%, the error message 'Total Power Exceeds Limit' will appear.

Temperature monitors mirror temperature. If the temperature exceeds 50°C, the actuators are set to 0mA so the mirror can cool down. A hardware temperature control can also be activated. If this is the case, mirror will have to be completely restarted (disconnect and reconnect both USB and power supply before restarting the mirror).

9.3 Message log

All error messages are recorded in the *Message log* box, with a timestamp indicating when the error appeared.

10 Using mu-DM with WaveTune

If you have purchased WaveTune, please read the following section.

Before installing WaveSuite, mu-DM Suite must have been correctly installed (see part 8).

Use the WaveSuite installer on the installation key provided with your wavefront sensor to install WaveSuite. During the installation process, a mirror configuration file will be needed. This configuration file is the one present in the WaveSuite installation folder: WFS_MuDM_XXXX.dat, with XXXX the mirror serial number.

If a patch for WaveTune is also available on the key, please install it as well.

All information about WaveTune software can be found in the software User Manual. The next few paragraphs detail information specific to the use of mu-DM with WaveTune software.

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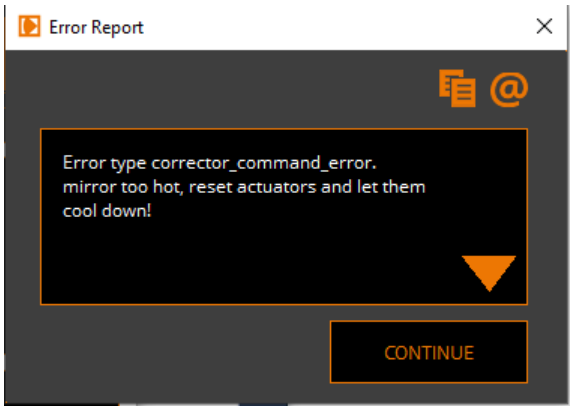
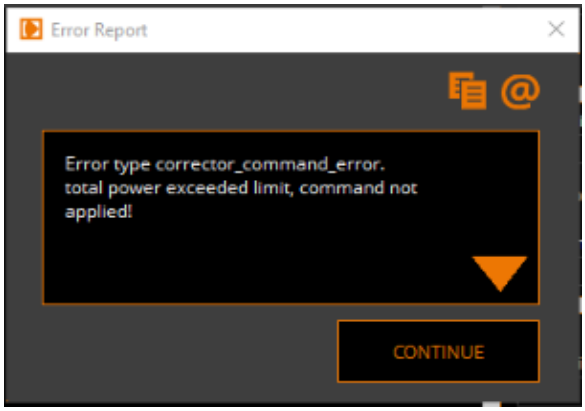
10.1 Error messages

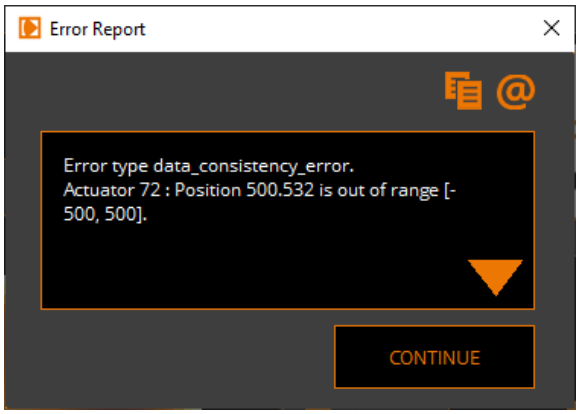
- if the temperature of the mirror is too high, the mirror will go into security mode. In this case, the following error message will appear, the mirror will be switched off and the software will have to be opened again to connect the mirror.



- if a shape sent to the mirror exceeds the limit in total power sent to the coils, the mirror will go into security mode. In this case, the mirror will not switch off, but will remain at the last command received before reaching the power limit. A software warning will appear



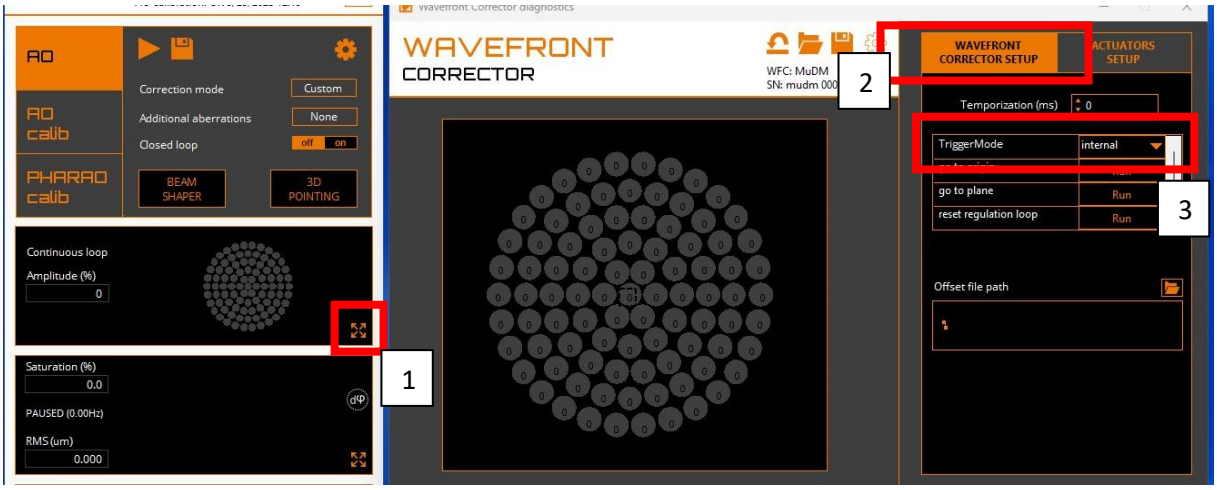
- If a command to apply a shape for which one of the actuators is over 100% of its dynamic range, the following error message appears, and the actuator is not moved. Each actuator can move over -500mA to 500mA



10.2 Trigger

To use the trigger function of the mirror, connect an SMB cable to “Trigger”. The signal sent to the trigger has to be a TTL signal. The mirror can be used in TRIG IN or TRIG OUT mode (but not both at the same time), but via WaveTune, only TRIG OUT is available. TRIG IN can only be used via the SDK.

To set the mirror to the desired trigger mode using WaveTune software: open the Wavefront Corrector Diagnostics panel (1), go to the Wavefront Corrector Setup tab in the advanced parameters (2), then choose the desired trigger mode using the ‘TriggerMode’ parameter (3).

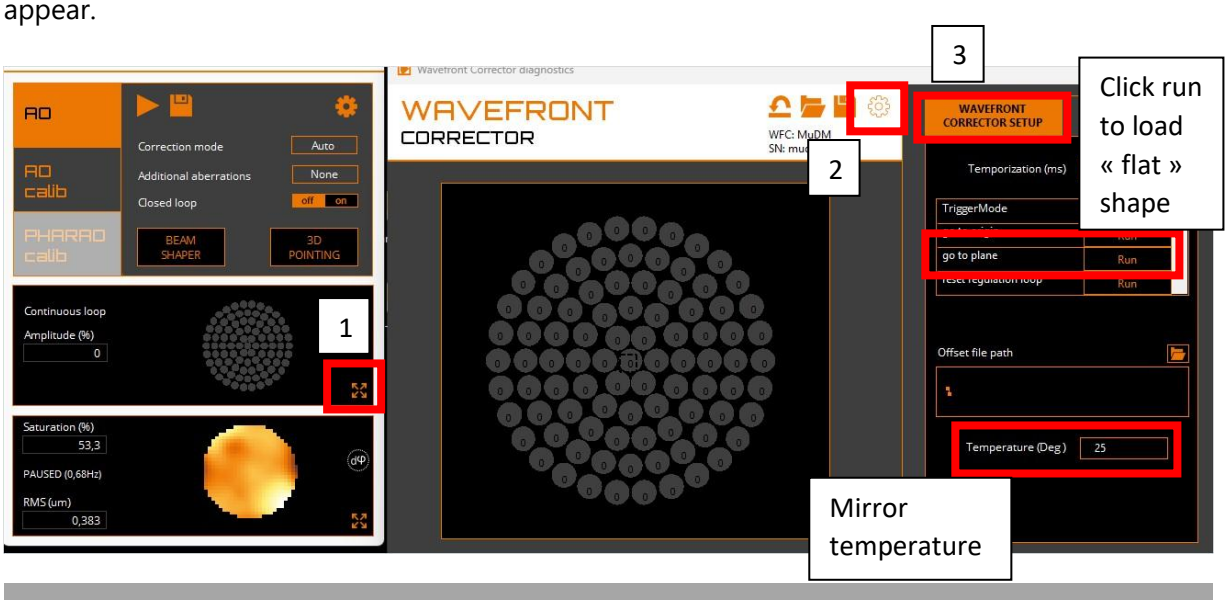


10.3 Full speed

To use the mirror at full speed, click on *AO Settings* (1), then choose Fast in Closed Loop Type (2). The mirror speed will be indicated next to the actuator layout on the main panel. Image refreshment rate is 1Hz by default, so in full speed it is not representative of the actual speed. This parameter can be changed in Fast Loop Display Rate in the settings of the AO settings panel (3).

10.4 Mirror temperature and flat

Mirror temperature will only appear when acquisition is paused. It might take a few seconds to appear.



11 Precautions

- Regulation option :
    - To ensure optimal stability, start using mu-DM only after at least 45min of it being connected to the software. After this warmup time, stability is ensured over 12h time even if environmental temperature is changing, unless environmental temperature changes too fast (more than 2°C in one hour).
    - After any larger change on the shape amplitude, to ensure optimal open loop stability, we recommend waiting until mirror temperature is stabilized.
    - Stability is not guaranteed for large mirror shapes.
- If mu-DM is used outside environmental temperature conditions, performance may be degraded.
- Do not pull on any of the cables while the mirror is in use.
- Do not disassemble mu-DM.
- Do not touch or clean the reflective surface of the mirror.
- Do not remove the protection window.

12 Support

For further questions, please contact our support team at **support@imagine-optic.com**.

Please keep the original packaging of the mirror, in case the mirror has to be shipped back.

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